

1 Training (simulation, n = 20)

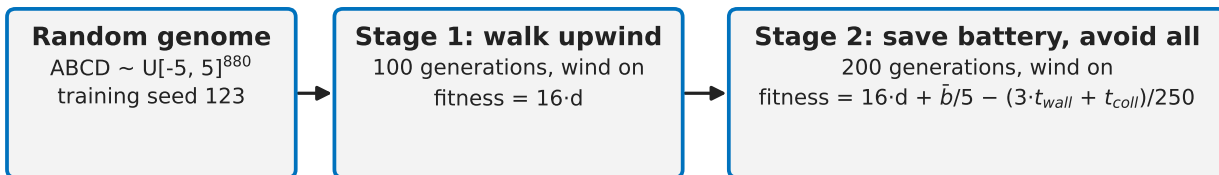
2 Evaluation & selection

3 Use

Unclamped variant — no safety mechanisms

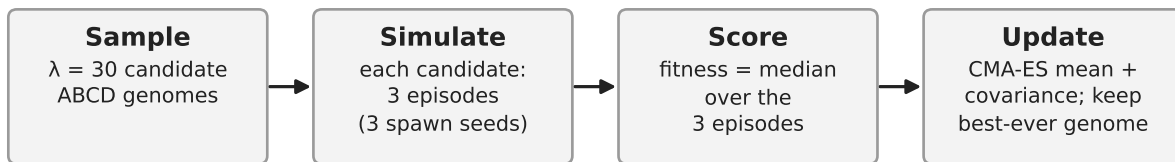


Clamped variant — speed clamp near robots/walls, 1.3× collision radius, soft collision resolution



Inside every stage: one CMA-ES generation

Stage 2 starts from Stage 1's best genome; the saved genome is the best ever evaluated, not the last one.



next generation

Evaluate

30 spawn seeds × every
swarm size n = 1...20
standard physics

vs. rule-based LJ
baseline (same seeds)
→ distance, battery,
both-beat rate

Select

5 training seeds per recipe;
candidates that reliably
beat LJ at n = 20, then
rejected any that hug
walls → plain_seed123

Simulation results

unclamped genome:
best simulated
performance

Hardware trial

clamped genome on
Thymio + Raspberry Pi

OptiTrack poses →
same virtual 4-quadrant
sensor as simulation

clamp re-enforced
on-robot